



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)
ALLEN COMMON TEST
05-09-2021

JEE(Main + Advanced) : NURTURE COURSE

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	D	B	C	A	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	C,D	A,B	B,C,D	A,B,C,D	B,D	B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	26.00	10.00	46.00	2.00	420.00	6.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	C	A	D	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	A,C,D	B,C,D	A,C	A,C,D	A,B
SECTION-II	Q.	1	2	3	4	5	6
	A.	6.00	2.50	4.00	80.00	5.00	5.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	B	B	B	A	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	B,C	C,D	A,B,C	B,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2011.00	2.00	7.00	4.00	2.00	-2.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (D)**

$$[F] = [M^1 L^1 T^{-2}], [F^0] = [M^1 L^0 T^{-2}]$$

$$\Rightarrow M^1 T^{-2} = 1 \Rightarrow T = M^{\frac{1}{2}}$$

(A) $[V] = [L^1 T^{-1}]$,

$$[V] = [M^0 L^0 M^{-\frac{1}{2}}] = M^{\frac{1}{2}}$$

(B) $[a] = [L T^{-2}]$,

$$[a] = [M^0 L^0 (M^{\frac{1}{2}})^{-2}] = M^{-1}$$

(C) $[P] = [M^1 L^2 T^{-3}]$,

$$[P] = [M^1 L^0 (M^{\frac{1}{2}})^{-3}] = M^{-\frac{1}{2}}$$

(D) $[L] = [M^1 L^2 T^{-1}]$,

$$[L] = [M^1 L^0 M^{-\frac{1}{2}}] = M^{\frac{1}{2}}$$

2. **Ans (D)**

If two vectors are perpendicular then $\vec{A} \cdot \vec{B} = 0$

$$(10\hat{i} - 2t\hat{j}) \cdot (5\hat{i} + 5\hat{j}) = 0 \Rightarrow 50 - 10t = 0 \Rightarrow t = 5 \text{ sec}$$

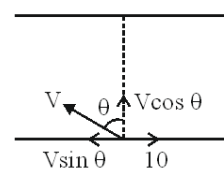
3. **Ans (B)**

As boy catches the ball, the ball is thrown perpendicular to inclined plane.

$$T = \frac{2u}{g \cos \theta}$$

4. **Ans (C)**

Let speed of ant relative to belt is V , if ant reaches at B



$$V \sin \theta = 10$$

$$V \cos \theta = \frac{50}{10} = 5$$

$$V = \sqrt{10^2 + 5^2} = 5\sqrt{5}$$

5. Ans (A)

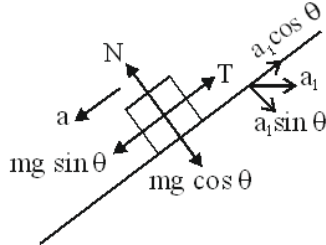
Let a = acceleration of m relative to the wedge A

a_1 = acceleration of wedge A towards right

a_2 = acceleration of Block B towards left

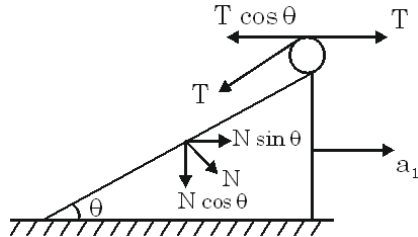
we can say

$$a = a_1 + a_2 \dots (1)$$

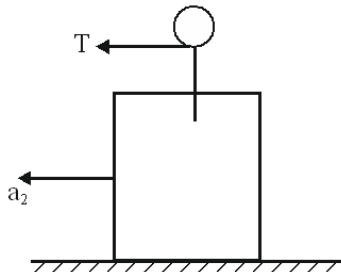


$$mg \cos \theta - N = ma_1 \sin \theta \dots (2)$$

$$mg \sin \theta - T = m(a - a_1 \cos \theta) \dots (3)$$



$$T - T \cos \theta + N \sin \theta = 2ma_1 \dots (4)$$



$$T = ma_2 \dots (5)$$

from equation (2) $N = mg \cos \theta - ma_1 \sin \theta$

using this in equation (4)

$$T(1 - \cos \theta) + (mg \cos \theta - ma_1 \sin \theta) \sin \theta = 2ma_1$$

$$\frac{T}{5} + \frac{12mg}{25} = \frac{59ma_1}{25} \text{ (as } \theta \text{ is } 37^\circ)$$

using equation (5) in above relation we get

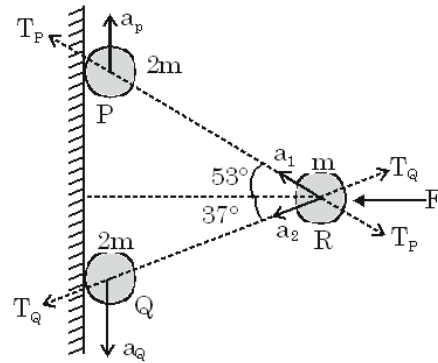
$$59a_1 - 5a_2 = 12g \dots (6)$$

solving equation (3) and (5) we get

$$5a_2 - 4a_1 + 5a = 3g \dots (7)$$

$$\text{solving equation (1), (6) \& (7) we get } a_1 = \frac{27g}{119}$$

6. Ans (C)



Let component of acceleration of disc R are

a_1 and a_2

$$a_1 = a_p \cos 37^\circ = \frac{4a_p}{5} \dots (1)$$

$$a_2 = a_Q \cos 53^\circ = \frac{3a_Q}{5} \dots (2)$$

for P

$$T_P \cos 37^\circ = 2ma_p$$

$$\frac{4T_P}{5} = 2ma_p \dots (3)$$

For Q

$$T_Q \cos 53^\circ = 2ma_Q$$

$$\frac{3T_Q}{5} = 2ma_Q \dots (4)$$

For R

$$F \cos 53^\circ - T_P = ma_1$$

$$\frac{3F}{5} - T_P = ma_1 \dots (5)$$

$$\text{and } F \cos 37^\circ - T_Q = ma_2$$

$$\frac{4F}{5} - T_Q = ma_2$$

from (3) & (5)

$$\frac{3F}{5} - \frac{10}{4}ma_p = ma_1$$

using (1) in above relation we get

$$a_p = \frac{2F}{11m}$$

SECTION-I (ii)

7. **Ans (C,D)**

$$v_{av} = \frac{\text{displacement}}{\text{time}}$$

as acceleration is constant $v_{av} = \frac{u+v}{2}$

$$\therefore \frac{u+v}{2} = \frac{6}{3} = 2$$

$$\frac{u+v}{2} = 2 \text{ m/s}$$

Hence u cannot be greater than 4 m/s

maximum retardation is $\frac{u}{t} = \frac{4}{3} \text{ m/s}^2$

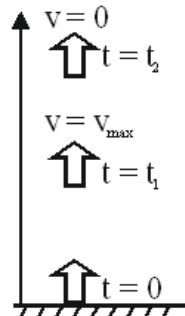
8. **Ans (A,B)**

Initially, acceleration is positive. It becomes zero at time

$$t_1 = \sqrt{\frac{g}{k}}$$

At this time velocity of rocket is maximum ($v_{\max}$)

For $t > t_1$ the acceleration becomes negative and rocket retards and finally it stops at ($t = t_2$)



$$\frac{dv}{dt} = g - kt^2$$

$$\int_0^v dv = g \int_0^t dt - k \int_0^t t^2 dt$$

$$v = gt - \frac{kt^3}{3} \dots (1)$$

If velocity becomes zero at time t then

$$t = t_2 = \sqrt{\frac{3g}{k}}$$

maximum velocity during upward motion is at

$$t = t_1 = \sqrt{\frac{g}{k}}$$

from equation (1)

$$v_{\max} = \frac{2}{3}g\sqrt{\frac{g}{k}}$$

for maximum height $v = \frac{dy}{dt} = gt - \frac{kt^3}{3}$

$$\int_0^{y_{\max}} dy = \int_0^{t_2} \left(gt - \frac{kt^3}{3} \right) dt; \left(t_2 = \sqrt{\frac{3g}{k}} \right)$$

$$\text{solving this } y_{\max} = \frac{3g^2}{4k}$$

time of fall from highest point is

$$t_3 = \sqrt{\frac{2y_{\max}}{g}} = \sqrt{\frac{3g}{2k}}$$

Total time of flight is

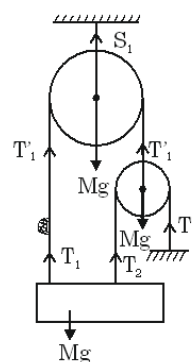
$$T = t_2 + t_3 = \sqrt{\frac{3g}{k}} \left(1 + \frac{1}{\sqrt{2}} \right)$$

9. **Ans (B,C,D)**

$$2T_2 + Mg = T'_1 \text{ (for pulley B) } \dots (1)$$

$$T'_1 = T_1 + \frac{Mg}{2} \text{ (at point D) } \dots (2)$$

$$T_1 + T_2 = Mg \text{ (for block C) } \dots (3)$$



adding equations (1), (2) & (3) we get

$$T_2 = \frac{Mg}{6} \dots (4)$$

from equation (1) & (4)

$$T'_1 = \frac{4Mg}{3}$$

tension in string S_1 is

$$Mg + 2T'_1 = \frac{11Mg}{3}$$

10. Ans (A,B,C,D)

If force applied by atmosphere on helicopter is

F then

$$(M + m)g - F = (M + m)a$$

$$F = (M + m)(g - a)$$

$$F = 11000(8) = 88000\text{N}$$

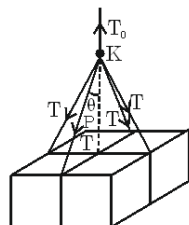
Let tension in string HK is T_0 and tension in

string AK, BK, CK and DK all are T

then for helicopter $Mg + T_0 - F = Ma$

solving this we get

$$T_0 = 8000$$



at point K $4T\cos\theta = T_0$

$$= 4T \frac{\sqrt{\ell^2 - \left(\frac{\ell}{2}\right)^2}}{\ell} = T_0$$

$$T = \frac{T_0}{2\sqrt{3}} = \frac{4000}{\sqrt{3}}$$

If string are connected to PQRS, θ will increase

$\cos\theta$ will decrease.

Hence T_0 and F will remain unchanged while T

will increase.

11. Ans (B,D)

Let the acceleration of wedge be 'a' down the incline.

Sum of masses of all rectangular blocks

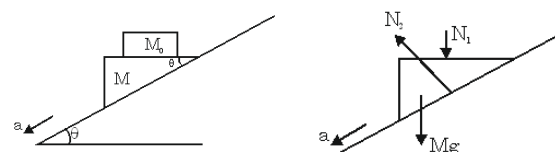
$$M_0 = M + \frac{M}{2} + \frac{M}{4} + \frac{M}{8} + \dots$$

$$= M \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right]$$

$$M_0 = M \frac{1}{\left(1 - \frac{1}{2}\right)} = 2M$$

For finding a we can treat the system as shown below.

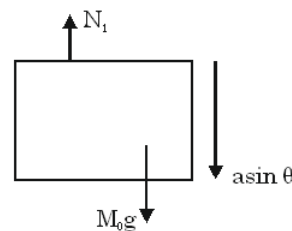
Acceleration of M_0 will be $a\sin\theta$ ($\downarrow$)



for wedge $N_2 = (N_1 + Mg)\cos\theta$ (1)

and $(N_1 + Mg)\sin\theta = Ma$ (2)

For M_0



$$M_0 g - N_1 = M_0 a\sin\theta$$

$$2Mg - N_1 = 2M a\sin\theta$$
(3)

Multiply (3) with $\sin\theta$ and add to (2)

$$3Mg \sin\theta = Ma(1 + 2\sin^2\theta)$$

$$\therefore a = \frac{3g \sin\theta}{1 + 2\sin^2\theta}$$

Acceleration of rectangular blocks

$$= a \sin\theta (\downarrow) = \frac{3g \sin^2\theta}{1 + 2\sin^2\theta} (\downarrow)$$

Mass of all rectangular blocks except 1 is M. Replace blocks 2, 3, 4, 5, with a single block of mass M considering motion of this mass M,

$$Mg - N_{12} = Ma\sin\theta$$

$$\therefore N_{12} = Mg - \frac{3Mg \sin^2\theta}{1 + 2\sin^2\theta} = \frac{Mg \cos^2\theta}{1 + 2\sin^2\theta}$$

12. **Ans (B,D)**

$$V_A = \sqrt{2g \sin \theta \times \frac{h}{\sin \theta}} = \sqrt{2gh}$$

$$V_B = \sqrt{2gh}$$

$$t_A = \sqrt{\frac{2h/\sin \theta}{g \sin \theta}} = \sqrt{\frac{2h}{g \sin^2 \theta}}$$

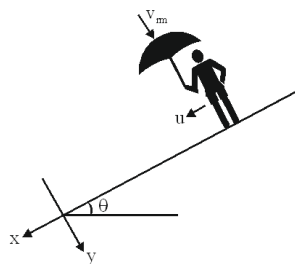
$$t_B = \sqrt{\frac{2h}{g}} \quad t_A > t_B$$

linear momentum $P = mv$ $P_B > P_A$

SECTION-II

1. **Ans (26.00)**

Velocity of rain relative to the man is perpendicular to the incline in this case (i.e. along the umbrella stick. This keeps umbrella perpendicular to the rainfall and provides maximum safety)



$$\vec{V}_{rm} = \vec{V}_r - \vec{V}_m$$

$$= (V_x \hat{i} + V_y \hat{j}) - u \hat{i} = (V_x - u) \hat{i} + V_y \hat{j}$$

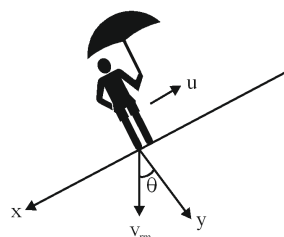
Since $\vec{V}_{rm}$ has no component

$$\therefore V_x = u$$

when the man is walking up, $\vec{V}_{rm}$ is directed vertically downward.

$$\vec{V}_{rm} = (V_x \hat{i} + V_y \hat{j}) - (-u \hat{j})$$

$$= u \hat{i} + V_y \hat{j} + u \hat{i} = 2u \hat{i} + V_y \hat{j}$$



$$\tan \theta = \frac{2u}{V_y}$$

$$\frac{1}{\sqrt{3}} = \frac{2u}{V_y} \Rightarrow V_y = 2\sqrt{3}u$$

$$\therefore V_r = \sqrt{V_x^2 + V_y^2} = \sqrt{13}u$$

2. **Ans (10.00)**

$$10\sqrt{3} \cos 60^\circ = v \cos 30^\circ$$

3. **Ans (46.00)**

Area between a-t graph and t-axis gives change in velocity.

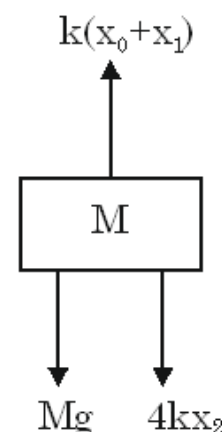
4. **Ans (2.00)**

Let initial extension in S_1 be x_0

$$kx_0 = Mg$$

Let extension in S_2 be x_2 and further extension in S_1 be x_1 as point A moves down by x .

$$2x_1 + x_2 = x \quad \dots\dots(1)$$



For equilibrium of M

$$k(x_0 + x_1) = Mg + 4kx_2$$

But $kx_0 = Mg$

$$\therefore x_1 = 4x_2 \quad \dots\dots(2)$$

Solving (1) and (2)

$$x_2 = \frac{x}{9}, \quad x_1 = \frac{4x}{9}$$

5. **Ans (420.00)**

Total impulse of all the force acting on the man is zero in 0 to 7s interval,

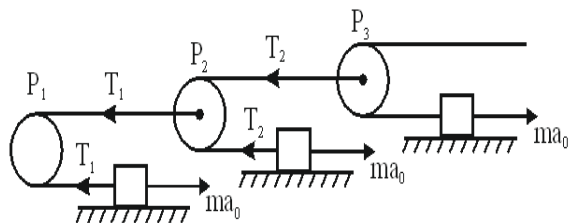
$\therefore$ area between F-t curve and t-axis is zero

$$\frac{1}{2} \times 200(3 + 2.4) - (600 - F_0) \times 3 = 0$$

$$F_0 = 420$$

6. **Ans (6.00)**

Let acceleration of pulley P_1 is $a_0 = 4\text{m/s}^2$
Consider the problem in a reference frame moving to left with an acceleration of a_0 . In this frame P_1 is at rest and all the blocks experience a force ma_0 towards right.



$$T_1 \propto a_0 \quad \dots(1)$$

In this frame let the acceleration of block 1 be $a_1(\leftarrow)$. Pulley P_2 will move towards right with an acceleration of a_1 . If an observer is located in reference frame attached to P_2 [which is moving to left with an acceleration of $(a_0 - a_1)$], he will similarly conclude that

$$T_2 \propto (a_0 - a_1) \quad \dots(2)$$

$$\frac{T_1}{T_2} = \frac{a_0}{a_0 - a_1}$$

But pulley is massless

$$\therefore T_1 = 2T_2$$

$$\therefore \frac{a_0}{a_0 - a_1} = 2$$

$$a_1 = \frac{a_0}{2}$$

In ground frame, acceleration of block 1 will be $\frac{3a_0}{2}(\leftarrow) = 6\text{m/s}^2$.

PART-2 : CHEMISTRY

SECTION-I (i)

4. **Ans (D)**

The anionic part of solid Cl_2O_6 is ClO_4^- .

6. **Ans (C)**

$$\text{V.D.} = \frac{M'}{2}, M' = 132$$

metal chloroide MCl_n

$$\Rightarrow M + n \times 35.5 = 132 \quad \dots (1)$$

Metal oxide M_2O_n

$$\Rightarrow \frac{2 \times M}{2M + 16n} \times 100 = 53 \quad \dots(2)$$

Solving (1) and (2)

$$M = 27$$

SECTION-I (ii)

7. **Ans (A,C,D)**

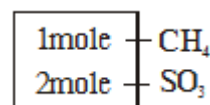
Moles of $\text{FeSO}_4 \cdot 6\text{H}_2\text{O}$

$$= \frac{2.6 \times 1000}{56 + 32 + 64 + 18 \times 6} = 10\text{moles}$$

8. **Ans (A,C,D)**

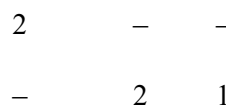
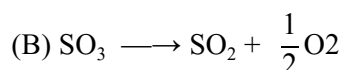
Due to the more cohesive energy in Li, it has highest melting point among alkali metals.

9. **Ans (B,C,D)**



$$\text{Avg. molar mass} = \frac{M_1x_1 + M_2x_2}{x_1 + x_2}$$

$$= \frac{16 + 80 \times 2}{3} = \frac{176}{3}$$



Avg. molar mass

$$= \frac{16 + 64 \times 2 + 32}{4} = \frac{176}{4}$$

$$\% \text{ change} = \frac{\frac{176}{3} - \frac{176}{4}}{\frac{176}{3}} \times 100 = 25\%$$

$$\text{(C) Mass ratio} = \frac{16}{2 \times 80} = \frac{1}{10}$$

$$\text{(D) Avg.} = \frac{2 \times 16 + 80 \times 2}{4} = \frac{192}{4} = 48$$

10. **Ans (A,C)**



↓

xgm

$\text{NaCl}(4.44-x)\text{ gm}$

$$\text{POAC on Ca} \quad \frac{x}{40 + 71} \times 1 = \frac{1.12 \times 1}{40 + 16}$$

$$= x = 2.22 \text{ gm}$$

11. **Ans (A,C,D)**

Let $W_{AB} = x$ gm so $W_{AC_2} = (400 - x)$ gm

So $n_{AB} = \frac{x}{40}$ mole & $n_{AC_2} = \frac{(400 - x)}{60}$ mole

$$W_A = \left(\frac{x}{40} + \frac{400 - x}{60} \right) \times 15 = \frac{26.25}{100} \times 400$$

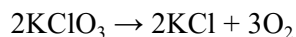
$$\Rightarrow x = 40 \text{ gm} = W_{AB} \text{ \& } W_{AC_2} = 360 \text{ gm}$$

$$\% \text{ mass of AB} = \frac{40}{400} \times 100 = 10\%$$

$$n_{AC_2} = \frac{360}{60} = 6 ; n_{AB} = \frac{40}{40} = 1$$

Total moles of A = 6 + 1 = 7

12. **Ans (A,B)**



$$\frac{490}{122.5} = 4 \text{ moles} \rightarrow \text{T.Y} = \frac{3 \times 4}{2} \text{ moles}$$

But % yield is 50%

$\therefore$ only 2 moles of KCl and 3 moles of O_2 produced

SECTION-II

2. **Ans (2.50)**

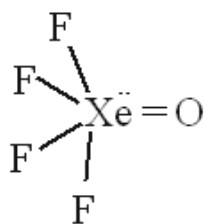
$$\text{No. of moles of } NH_3 \text{ in 100g mixture} = \frac{68}{17} = 4$$

$$\text{No. of moles of } H_2 \text{ in 100 g mixture} = \frac{32}{2} = 16$$

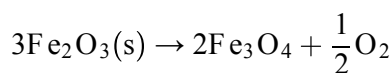
$$M_{\text{average}} = \frac{\text{Total mass}}{\text{Total moles}} = \frac{100}{4 + 16} = 5$$

$$V D = \frac{5}{2} = 2.5$$

3. **Ans (4.00)**



4. **Ans (80.00)**



$$3 \times 160 \qquad \qquad \frac{1}{2} \times 32$$

$$= 480 \text{ gm} \qquad \qquad = 16 \text{ gm}$$

loss of 16 gm $\rightarrow$ 480 gm Fe_2O_3

$$\text{loss of 0.04 gm} \rightarrow 0.04 \times \frac{480}{16} = 1.2 \text{ gm } Fe_2O_3$$

$$\% \text{ by mass} = \frac{1.2}{1.5} \times 100 = 80\%$$

6. **Ans (5.00)**

ClF_3 , XeF_2 , SF_4 , $[SbF_4]^-$, $XeOF_2$ are having lone pairs at equatorial position.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{2020}$$

$$\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{2020} \right) - 2$$

$$\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2020} \right)$$

$$\frac{1}{1011} + \frac{1}{1012} + \dots + \frac{1}{2020}$$

2. **Ans (B)**

We have

$$a_n = \frac{1}{n+1} \sum_{k=1}^n \left(\frac{1}{k} + \frac{1}{n+1-k} \right)$$

$$= \frac{2}{n+1} \sum_{k=1}^n \frac{1}{k}$$

For $n \geq 2$

$$\frac{1}{2}(a_n - a_{n+1}) = \frac{1}{n+1} \sum_{k=1}^n \frac{1}{k} - \frac{1}{n+2} \sum_{k=1}^{n+1} \frac{1}{k}$$

$$= \left(\frac{1}{n+1} - \frac{1}{n+2} \right) \sum_{k=1}^n \frac{1}{k} - \frac{1}{(n+1)(n+2)}$$

$$= \frac{1}{(n+1)(n+2)} \sum_{k=2}^n \frac{1}{k} > 0$$

$$\Rightarrow a_n > a_{n+1}$$

3. **Ans (B)**

RHS when simplified is equal x.

$$x_1 = 1; x_2 = \frac{1}{10}; x_3 = \frac{1}{100}$$

4. **Ans (B)**

$$T_r = \frac{1}{3} \left(\frac{(3r+1)^2 - (3r-2)^2}{(3r+1)^2(3r-2)^2} \right)$$

$$T_r = \frac{1}{3} \left(\frac{1}{(3r-2)^2} - \frac{1}{(3r+1)^2} \right)$$

$$\boxed{S_\infty = \frac{1}{3}}$$

5. **Ans (A)**

$$t_r = \frac{2r+5}{(r+1)(r+2)} \left(\frac{1}{3} \right)^r$$

$$= \frac{3(r+2) - (r+1)}{(r+1)(r+2)} \left(\frac{1}{3} \right)^r$$

$$= \frac{1}{r+1} \left(\frac{1}{3} \right)^{r-1} - \frac{1}{r+2} \left(\frac{1}{3} \right)^r$$

$$\sum_{r=1}^{10} t_r = \frac{1}{2} - \frac{1}{12} \times \frac{1}{3^{10}}$$

6. **Ans (C)**

$$\text{Let } \sqrt{768} = 32 \cos \theta$$

$$16\sqrt{3} = 32 \cos \theta$$

$$\cos \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$$

$$= \sqrt{4 + \sqrt{8 - 8 \cos \frac{\pi}{12}}}$$

$$= \sqrt{4 + 4 \sin \frac{\pi}{24}} = \sqrt{4 + 4 \cos \frac{11\pi}{24}}$$

$$= \sqrt{4 \left(1 + \cos \frac{11\pi}{24} \right)} = \sqrt{4.2 \cos^2 \frac{11\pi}{48}} = 2\sqrt{2} \cos \frac{11\pi}{48}$$

SECTION-I (ii)

7. **Ans (A,C,D)**

$$f(x) = 1 - \cos^2(x + \alpha) + \sin^2(x + \beta) - 2\cos(\alpha - \beta)$$

$$\sin(x + \alpha) \sin(x + \beta)$$

$$= 1 - \cos(2x + \alpha + \beta) \cos(\alpha - \beta) - 2\cos(\alpha - \beta)$$

$$\sin(x + \alpha) \sin(x + \beta)$$

$$= 1 - \cos(\alpha - \beta) [\cos(2x + \alpha + \beta) - \cos(2x + \alpha + \beta)]$$

$$+ \cos(\alpha - \beta)]$$

$$= \sin^2(\alpha - \beta)$$

a constant function

8. **Ans (B,C)**

$$f(n, \theta) =$$

$$\sin \theta \left(\frac{1 - \cos \theta}{2} \right) + \frac{\sin 2\theta}{2} \left(\frac{1 - \cos 2\theta}{2} \right) + \frac{\sin 4\theta}{4} \left(\frac{1 - \cos 4\theta}{2} \right) + \dots + \frac{\sin 2^n \theta}{2^n} \left(\frac{1 - \cos 2^n \theta}{2} \right)$$

$$f(n, \theta) = \frac{1}{2} \left[\sin \theta - \frac{\sin 2\theta}{2} + \frac{\sin 2\theta}{2} - \frac{\sin 4\theta}{4} + \dots - \frac{\sin 2^{n+1} \theta}{2^{n+1}} \right]$$

$$f(n, \theta) = \frac{1}{2} \left[\sin \theta - \frac{\sin 2^{n+1} \theta}{2^{n+1}} \right]$$

9. **Ans (C,D)**

If a, b, c are p^{th} , q^{th} and r^{th} terms of an A.P. then

$\frac{b-c}{a-b}$ is always a rational number.

10. **Ans (A,B,C)**

$$P + Q = \sum_{k=0}^{100} k (\sin^2 \theta + \cos^2 \theta) = \sum_{k=0}^{100} k = 5050$$

11. **Ans (B,C,D)**

$$\text{Let } f(x) = ax^2 + bx + c \text{ and } f(1) = a + b + c < 0$$

$$\Rightarrow f(x) < 0 \text{ for all } x \in \mathbb{R}$$

$$\Rightarrow f(-2) < 0$$

$$\Rightarrow 4a - 2b + c < 0$$

$$\text{Again } f(3) + f(2) < 0$$

$$\Rightarrow 13a + 5b + 2c < 0$$

$$\text{Also } f(-5) < 0 \Rightarrow 25a - 5b + c < 0$$

$$\Rightarrow 5b - 25a - c > 0$$

12. **Ans (A,B,D)**

$$(a+b) = x, \text{ then } x^3 - 21x + 20 = 0$$

Which has roots 1, 4, -5

SECTION-II

1. **Ans (2011.00)**

$$T_n = \sqrt{1 + \frac{1}{n^2} + \frac{1}{(n+1)^2}} = \sqrt{\frac{n^4 + 2n^3 + n^2 + n^2 + 2n + 1 + n^2}{n^2(n+1)^2}} = \frac{n^2 + n + 1}{n(n+1)} = 1 + \frac{1}{n(n+1)}$$

2. **Ans (2.00)**

$$x^2 + (a-b)x + (1-a-b) = 0$$

$$\therefore D > 0$$

$$\Rightarrow (a-b)^2 - 4 \times 1 \times (1-a-b) > 0$$

$$a^2 + b^2 - 2ab - 4 + 4a + 4b > 0$$

$$\therefore b^2 + 2b(2-a) + (a^2 + 4a - 4) > 0$$

$$\therefore 4(2-a)^2 - 4 \times 1 \times (a^2 + 4a - 4) < 0$$

$$4 + a^2 - 4a - a^2 - 4a + 4 < 0$$

$$8a - 8 > 0$$

$$a > 1$$

3. **Ans (7.00)**

$$z^2y^2 + y^2 + x^2 + 10 = 6x + 6y$$

$$(xy-1)^2 + (x+y-3)^2 = 0$$

$$x+y=3$$

$$xy=1$$

$$x^2 + y^2 + 2 = 9$$

$$x^2 + y^2 = 7$$

4. **Ans (4.00)**

$\therefore$ sum of two A.P.'s also an A.P.

$$x_4 + y_4 + z_4 + w_4 = A + 3D = 8 \text{ \&}$$

$$x_{10} + y_{10} + z_{10} + w_{10} = A + 9D = 20$$

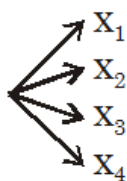
$$\Rightarrow A = 2, D = 2$$

$$\Rightarrow x_{20} + y_{20} + z_{20} + w_{20} = A + 19D = 40$$

$$\text{Now } \frac{x_{20} + y_{20} + z_{20} + w_{20}}{4} \geq (x_{20} \cdot y_{20} \cdot z_{20} \cdot w_{20})^{1/4}$$

$$\Rightarrow (x_{20} \cdot y_{20} \cdot z_{20} \cdot w_{20}) \leq 10^4$$

5. **Ans (2.00)**

$$x^4 - 4x^3 + ax^2 - bx + 1 = 0$$


$$\sum x_i = 4$$

$$\prod x_i = 1$$

$\therefore$ A.M. of x_i = G.M. of x_i

$$x_1 = x_2 = x_3 = x_4 = 1$$

6. **Ans (-2.00)**

$$a_1 = 2$$

$$a_2 = 1 - \frac{1}{2} = \frac{1}{2}$$

$$a_3 = 1 - 2 = -1$$

$$a_4 = 1 + 1 = 2 \text{ \& so on}$$

$$a_1 = a_4 = a_7 = \dots = 2$$

$$a_2 = a_5 = a_8 = \dots = \frac{1}{2}$$

$$a_3 = a_6 = a_9 = \dots = -1$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

ACT

05-09-2021

JEE(Main + Advanced) : Nurture Course (PHASE : ALLEN COMMON TEST)

Test Type : ACT

Test Pattern : JEE (Main)

TEST DATE : 05-09-2021

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	B	D	B	A	B	B	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	B	B	D	B	B	C	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
A.	13.39 to 13.50	4.00	0.25	64.00	0.66 to 0.67	5000.00	0.60	50.00	2.50	25.00	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	D	C	D	D	C	D	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	A	C	A	B	B	B	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
A.	5.00	5.00	7.00	33.33	5.00	6.00	12.50	10.40	9.00	6.00	

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	A	D	C	B	B	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	B	A	C	B	C	B	A	D	D
	Q.	1	2	3	4	5	6	7	8	9	10
A.	2.00	3.00	1.00	1.00	3.00	7.00	7.00	2.00	4.00	4.00	

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)

$$F = 600 - 2 \times 10^5 t$$

$$F = 0, t = \frac{600}{20 \times 10^5} = 3 \times 10^{-3} \text{ sec}$$

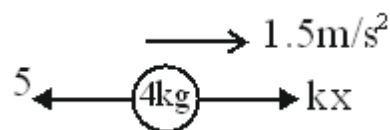
$$I = \int_0^t F \cdot dt = \int_0^t (600 - 2 \times 10^5 t) dt$$

$$I = [600t - 10^5 \cdot t^2]_0^t$$

$$I = 600 \times 3 \times 10^{-3} - 10^5 \times 9 \times 10^{-6}$$

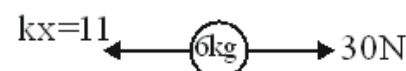
$$1.8 - 0.9 = 0.9 \text{ N-S}$$

2. Ans (C)



$$kx - 5 = 4 \times 1.5$$

$$kx = 11$$

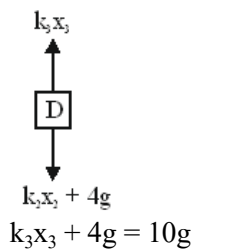
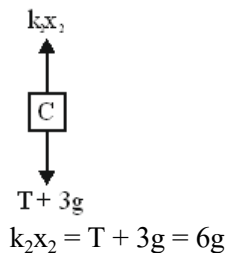
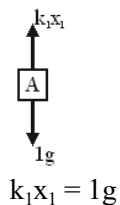
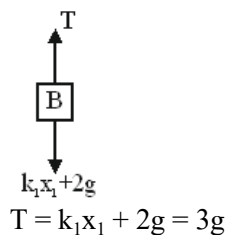


$$a = \frac{30 - 11}{6} = \frac{19}{6}$$

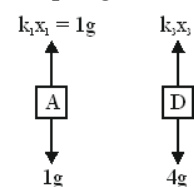
$$= 3.17 \text{ m/s}^2$$

3. **Ans (B)**

Before string is cut

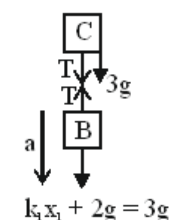


when spring between D and C is cut $k_2x_2 = 0$
as spring force does not changes instantaneously



$$a_D = \frac{k_3x_3 - 4g}{4} = \frac{3g}{2} (\uparrow)$$

$$a_A = 0$$



$$T + 3g = 3a \quad \dots (1) \quad (\text{for block C})$$

$$3g - T = 2a \quad \dots (2) \quad (\text{for block B})$$

$$(1) + (2)$$

$$6g = 5a$$

$$a = \frac{5g}{6} (\downarrow)$$

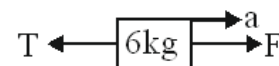
5. **Ans (B)**

Let acceleration of 6kg block is a

$$v^2 = 2as$$

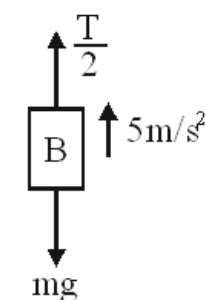
$$(5)^2 = 2as, a = \frac{5}{2} \text{ m/s}^2$$

$$a_B = 2a_A = 5 \text{ m/s}^2$$



$$F - T = 6a$$

$$75 - T = 6 \times \frac{5}{2}, T = 60\text{N}$$



$$\frac{T}{2} - mg = ma$$

$$30 - m \times 10 = m \times 5$$

$$m = 2\text{kg}$$

6. **Ans (A)**

$$\vec{a} = \frac{\vec{F}}{m} = -6\hat{i} - 5\hat{j}$$

$$v_x = u_x + a_x t$$

$$0 = 30 - 6t \Rightarrow t = 5 \text{ sec}$$

7. **Ans (B)**

$$R = \frac{2u^2 \sin \theta \cos \theta}{g} = 38.4\text{m}$$

$$T = \frac{2u \sin \theta}{g} = 2.4\text{s}$$

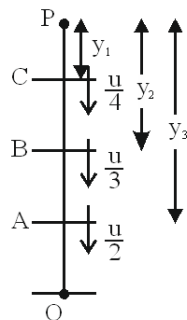
man has to travel $38.4 - 7.2 = 31.2\text{m}$ in 2.4s

$$\text{so } v_{\text{man}} = \frac{31.2}{2.4} = 13\text{m/s}$$

8. **Ans (B)**

$$\text{At maximum } x \text{ coordinate } v = \frac{dx}{dt} = 0$$

9. **Ans (D)**



$$\frac{u}{2} = \sqrt{2gy_3}$$

$$y_3 = \frac{v^2}{8g}$$

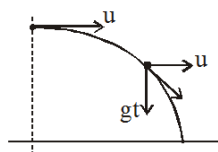
in similar way

$$y_2 = \frac{v^2}{18g}$$

$$y_1 = \frac{v^2}{32g}$$

$$\frac{AB}{AC} = \frac{y_3 - y_2}{y_3 - y_1} = \frac{20}{27}$$

10. **Ans (B)**



$$\vec{v} = u\hat{i} + gt(-\hat{j})$$

$$\vec{v} = u\hat{i} + g(3)(-\hat{j})$$

$$|\vec{v}| = 2u$$

$$u^2 + 9g^2 = 4u^2$$

$$9g^2 = 3u^2$$

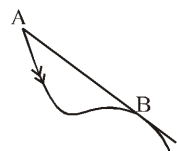
$$u = \sqrt{3}g = 10\sqrt{3} \text{ m/s}$$

11. **Ans (D)**

All blocks are having an acceleration of $g \sin \theta$ so contact force between any two blocks is zero.

12. **Ans (B)**

From A draw tangent to curve as shown



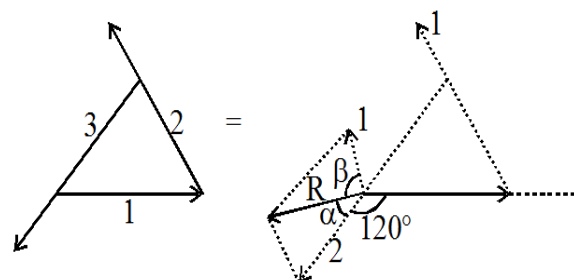
$\vec{AB}$ = total displacement vector

$\therefore \vec{AB}$ = same direction as average velocity

Also $\vec{AB}$ is tangent of both as B

$\Rightarrow \vec{AB}$ = same direction as velocity as B

13. **Ans (B)**



$$R = \sqrt{1^2 + 2^2 + 2 \cdot 1 \cdot 2 \cdot \cos 120^\circ} = \sqrt{3}$$

$$\tan \alpha = \frac{1 \sin 120^\circ}{2 + 1 \cos 120^\circ} = \frac{\sqrt{3}/2}{2 - 1/2} = \frac{1}{\sqrt{3}}$$

$$\alpha = 30^\circ$$

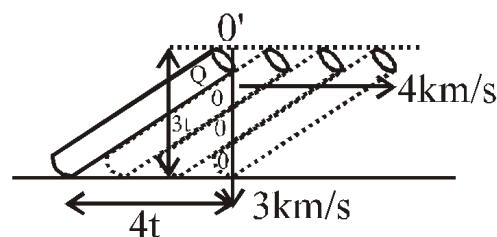
$\therefore 120^\circ + 30^\circ = 150^\circ$ from direction of 1 unit

$$\text{or } \tan \beta = \frac{2 \sin 120^\circ}{1 + 2 \cos 120^\circ} = \frac{2 \times \sqrt{3}/2}{1 - 1} = \infty$$

$$\beta = 90^\circ$$

$\therefore 90^\circ$ from direction of 2 unit vector.

14. **Ans (B)**



$$\tan \theta = \frac{4}{3}$$

$$\theta = 53^\circ$$

15. **Ans (D)**

$$\vec{A} \times (\vec{B} \times \vec{A}) = \vec{B}(\vec{A} \cdot \vec{A}) - \vec{A}(\vec{B} \cdot \vec{A})$$

$$\vec{B} = \frac{\vec{A} \times (\vec{B} \times \vec{A})}{\vec{A} \cdot \vec{A}} + \frac{\vec{A}(\vec{B} \cdot \vec{A})}{\vec{A} \cdot \vec{A}}$$

16. **Ans (B)**

$$[\lambda g] = L L T^{-2} = L^2 T^{-2} = [V^2]$$

17. **Ans (B)**

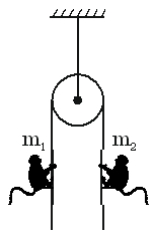
$$[P] = [M^0 L^1 T^0]^x [M^1 L^{-3} T^0]^y [M L^{-2} T^{-2}]^z$$

$$[M^1 L^2 T^{-3}] = [M^{x+y} L^{x-3y-2z} T^{-2z}]$$

$$-2z = -3, z = \frac{3}{2}$$

$$y + z = 1, y = -\frac{1}{2}$$

18. Ans (C)



let $m_1 > m_2$

(A) If m_2 is at rest $T = m_2g$

$m_1g - T = m_1a$ (when m_1 slides down)

$T = m_1(g - a)$ possible

(B) If m_1 is at rest $T = m_1g$

$T - m_2g = m_2a$ (when m_2 goes up)

$T = m_2(g + a)$ possible

(C) $m_1g - T = m_1a \Rightarrow T = m_1(g - a)$

$m_2g - T = m_2a \Rightarrow T = m_2(g - a)$

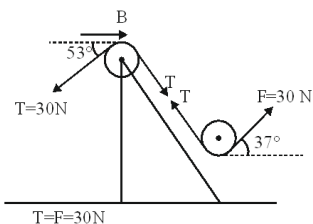
This is not possible

(D) $m_1g - T = m_1a$, $T = m_1(g - a)$

$T - m_2g = m_2a$, $T = m_2(g + a)$

This may be possible

19. Ans (C)



$$F \cos 37^\circ - T \cos 53^\circ = 5a$$

$$\Rightarrow a = 1.2 \text{ m/s}^2$$

20. Ans (C)

Let mass of ball is m and that of rod is M , and acceleration of ball is a then acceleration of rod is $2a$

$$\text{For rod } Mg - T = M(2a) \dots (1)$$

$$\text{For ball } 2T - mg = ma \dots (2)$$

Multiplying equation-(1) by 2 and adding to equation-(2)

$$2Mg - mg = (4M + m)a$$

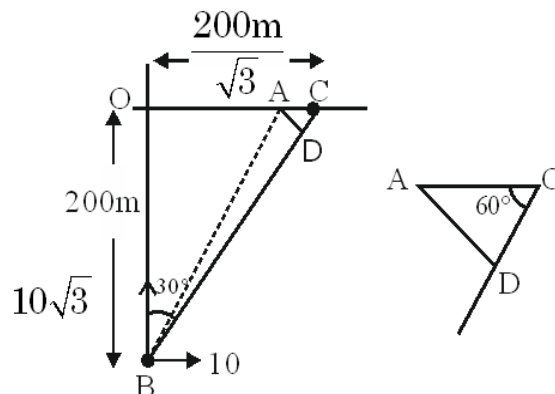
$$\text{or } a = \frac{2Mg - mg}{4M + m}$$

as $M = 1.5m$, $a = \frac{2g}{7}$, then relative acceleration of ball w.r.t. rod is $3a$ then time taken to cross is given

$$\text{by } t = \sqrt{\frac{2\ell}{a_{\text{rel}}}} = \sqrt{\frac{2\ell}{3a}}$$

SECTION-II

1. Ans (13.39 to 13.50)



Consider A at rest now w.r.t. A, B is moving at

angle 30° with OB

from $\triangle BOC$

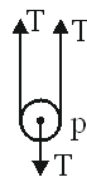
$$\tan 30^\circ = \frac{OC}{OB}, OC = \frac{200}{\sqrt{3}} \text{ and}$$

$$AC = \frac{200}{\sqrt{3}} - 100$$

from $\triangle ADC$

$$\sin 60^\circ = \frac{AD}{AC}, AD = \frac{\sqrt{3}}{2} \left(\frac{200}{\sqrt{3}} - 100 \right)$$

2. Ans (4.00)



$$2T - T = 0$$

$$T = 0$$

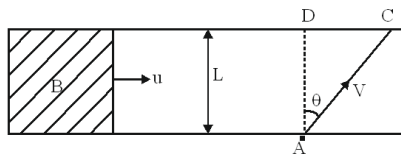
As pulleys are massless tension in string is zero

Acceleration of both block is $= g = 10 \text{ m/s}^2$

Acceleration of pulley p is $= 40 \text{ m/s}^2$

3. **Ans (0.25)**

Let the insect move with speed v along the direction shown in the figure.



$$\text{Time to cross is } t = \frac{L}{v \cos \theta}$$

$$DC = v \sin \theta \cdot t = L \frac{\sin \theta}{\cos \theta} = L \tan \theta$$

The block will just miss the insect if distance travelled by the block in time t is just less than $\sqrt{15}L + DC$

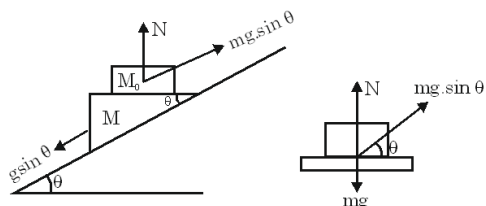
$$\Rightarrow u \cdot t = \sqrt{15}L + L \tan \theta$$

$$\Rightarrow u \cdot \frac{L}{v \cos \theta} = \sqrt{15}L + L \tan \theta$$

$$\Rightarrow v = \frac{uL}{\sqrt{15}L \cos \theta + L \sin \theta}$$

$$V_{\min} = \frac{uL}{4L} = \frac{u}{4}$$

4. **Ans (64.00)**

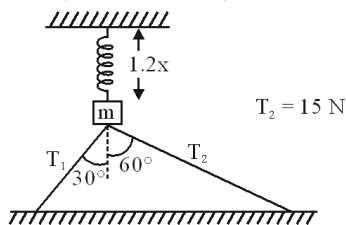


$$N + mg \sin \theta \cdot \sin \theta = mg$$

$$N = mg \cos^2 \theta$$

$$N = 10 \times 10 \times \frac{4 \times 4}{5 \times 5} = 64 \text{ newton}$$

5. **Ans (0.66 to 0.67)**



Along horizontal direction

$$T_1 \sin 30^\circ = T_2 \sin 60^\circ$$

$$\text{or } T_1 = \sqrt{3}T_2 \quad \dots(1)$$

Along vertical direction

$$k(0.2x) = mg + T_1 \cos 30^\circ + T_2 \cos 60^\circ$$

$$600(0.2x) = 5g + \sqrt{3}T_2 \times \frac{\sqrt{3}}{2} + \frac{T_2}{2}$$

$$120x = 50 + 2T_2$$

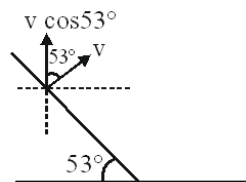
6. **Ans (5000.00)**

Using $N_1 U_1 = N_2 U_2$

$$N_2 = 1 \left[\left(\frac{M_1}{M_2} \right)^1 \left(\frac{L_1}{L_2} \right)^1 \left(\frac{T_1}{T_2} \right)^{-2} \right]$$

$$N_2 = 1 \left[\left(\frac{1\text{kg}}{0.1\text{kg}} \right)^1 \left(\frac{1\text{m}}{0.2\text{m}} \right)^1 \left(\frac{1\text{s}}{10\text{s}} \right)^{-2} \right] = 5000$$

7. **Ans (0.60)**



time taken to reach highest point

$$t = \frac{v \cos 53^\circ}{g} = 0.6\text{s}$$

8. **Ans (50.00)**

$$y = \frac{1}{2}gt^2, v_x = \frac{3}{8} \times \frac{1}{2}gt^2$$

when particle strikes to ground

$$v_y = \sqrt{2gH} = 40\text{m/s}$$

$$\text{time of flight} = \sqrt{\frac{2H}{g}} = 4\text{s}$$

$$v_x = \frac{3g}{16}t^2$$

At $t = 4\text{s}$

$$v_x = 3g = 30\text{m/s}$$

$$v = \sqrt{v_x^2 + v_y^2}$$

$$v = 50 \text{ m/s}$$

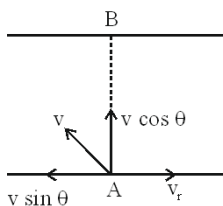
9. **Ans (2.50)**

$$3T - 200 \times 10 \times \frac{1}{2} = 200 a$$

$$3 \times 500 - 1000 = 200 a$$

$$a = 2.5 \text{ m/s}^2$$

10. **Ans (25.00)**



$$v \sin \theta = v_r = 7$$

$$v \cos \theta = \frac{120}{5} = 24$$

$$v = \sqrt{7^2 + 24^2} = 25\text{m/s}$$

17. **Ans (B)**

E.N. Difference of $O - X > O - H$
so HOX will dissociate in $HO^- + X^+$
hence nature of HOX will be basic.

18. **Ans (B)**

$$Ca \rightarrow \frac{20}{40} = \frac{1}{2} = 1$$

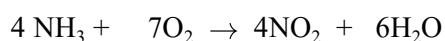
$$Br \rightarrow \frac{80}{80} = 1$$



19. **Ans (C)**

When reaction is reversed, magnitude of enthalpy change remains the same.

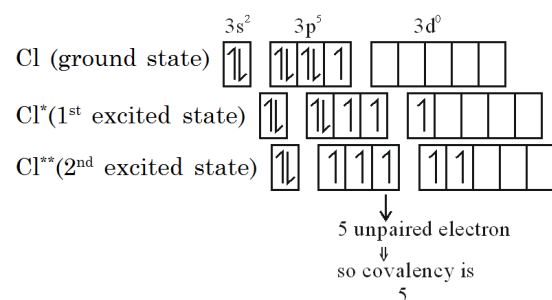
20. **Ans (C)**



$$\begin{array}{ccccccc} 4 \text{ mol} & 8 \text{ mol} & 4 \text{ mol} & 6 \text{ mol} & & & \\ & & & & 184 \text{ g} & 108 \text{ g} & \end{array}$$

SECTION-II

1. **Ans (5.00)**

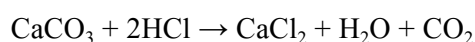


2. **Ans (5.00)**

Planar- BF_3 , BI_3 , CO_2 , $BeCl_2$, SO_3

Nonplanar $\ddot{N} (CH_3)_3$, CCl_4 , $CHCl_3$

4. **Ans (33.33)**



$$\text{Moles of HCl} = \frac{17.52}{36.5}$$

$$\text{Moles of } CaCO_3 = \frac{1}{2} \times \frac{17.52}{36.5}$$

$$\text{Weight of } CaCO_3 \text{ required} = \frac{1}{2} \times \frac{17.52}{36.5} \times 100$$

Mass of sample taken :

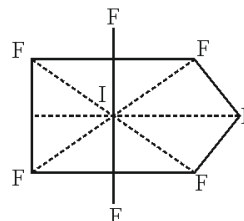
$$= \frac{1}{2} \times \frac{17.52}{36.5} \times \frac{100 \times 100}{72} = 33.33 \text{ gm}$$

5. **Ans (5.00)**

Bonding AsH_3 , PH_3 & H_2Se can not be explained by hybridisation as per Drago's rule.

6. **Ans (6.00)**

Hybridisation of $IF_7 = sp^3d^3$



7. **Ans (12.50)**

Given conditions-

$$P = 10^{-10} \text{ mm Hg} = \frac{3.8 \times 10^{-10}}{760} \text{ atm}$$

$$V = 1 \text{ cm}^3 = \frac{1}{1000} \text{ L}, T = 27^\circ\text{C} = 300 \text{ K}$$

No. of molecules, $N = ?$

Now, applying

$$pV = nRT$$

$\therefore$ number of molecules per cm^3 ,

$$N = \frac{N_A \times P \times V}{RT}$$

$$N = \frac{6 \times 10^{23} \times (3.8 \times 10^{-10} / 760) \times (1/1000)}{(0.08)(300)}$$

$$N = 12.50 \times 10^6 \text{ molecules}$$

8. **Ans (10.40)**

Mass of Li = 140 mg

$$= 140 \times 10^{-3} \text{ g}$$

$$= 14 \times 10^{-2} \text{ g}$$

$$\text{number of moles of Li} = \frac{14 \times 10^{-2}}{7} = 2 \times 10^{-2}$$

so energy req. to convert 2×10^{-2} mole atoms of lithium into Li^+ ion

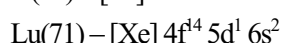
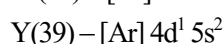
$$= 520 \times 2 \times 10^{-2} = 10.4 \text{ kJ}$$

9. **Ans (9.00)**

All are the correct order of 1st ionisation energy.

10. **Ans (6.00)**

Group 3



Sum of electron in outermost subshell = $2 + 2 + 2 = 6$.

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (C)**

$$S - S_n < \frac{1}{300}$$

$$\Rightarrow \frac{1}{1 - \frac{1}{3}} - \frac{1 - \frac{1}{3^n}}{1 - \frac{1}{3}} < \frac{1}{300}$$

$$\Rightarrow \frac{3}{2} \left[1 - 1 + \frac{1}{3^n} \right] < \frac{1}{300}$$

$$\Rightarrow \frac{1}{3^n} < \frac{1}{300} \times \frac{2}{3}$$

$$\Rightarrow \frac{1}{3^n} < \frac{1}{450} \Rightarrow 3^n > 450$$

$\therefore$ Least value of $n = 6$

2. **Ans (C)**

We have $a = b + \frac{1}{a + \frac{1}{a}}$

$$\Rightarrow (a - b) = \frac{1}{a + \frac{1}{a}} \quad \text{and} \quad b = a - \frac{1}{b + \frac{1}{b}}$$

$$\Rightarrow a - b = \frac{1}{b + \frac{1}{b}}$$

$$\therefore a + \frac{1}{a} = b + \frac{1}{b} \Rightarrow (ab - 1)(a - b) = 0$$

$$\therefore ab = 1 \quad \because a \neq b$$

$$\therefore (a - b) = \frac{1}{a + b} \Rightarrow a^2 - b^2 = 1$$

3. **Ans (B)**

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B} = 1$$

$$\Rightarrow A + B = \frac{\pi}{4}$$

4. **Ans (A)**

$2^0, (2^1, 2^1) (2^2, 2^2, 2^2, 2^2) \dots (2^k, 2^k, \dots, 2^k)$
 (2 time) (2² times) (2^k times)

2^k starts with 2^k th term and end with $(2^{k+1} - 1)$ th

2^{10} starts with 1024th terms and ends with 2047th terms

5. **Ans (D)**

$$2x^3 - 5x^2 - 7x + 3 = 0 \quad \begin{cases} \alpha \\ \beta \\ \gamma \end{cases}$$

$$\text{H.M.} = \frac{3}{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}} = \frac{3\alpha\beta\gamma}{\alpha\beta + \beta\gamma + \gamma\alpha} = \frac{3(-\frac{3}{2})}{(-\frac{7}{2})} = \frac{9}{7}$$

6. **Ans (C)**

$$\frac{2 \log 5 + \log 7}{\log 5 + 2 \log 7} = a, \quad \frac{3 \log 5 + \log 7}{\log 5 + 3 \log 7} = b$$

$$\frac{1 - ab}{a - b}$$

$$= \frac{(\log 5 + 2 \log 7)(\log 5 + 3 \log 7) - (2 \log 5 + \log 7)(3 \log 5 + \log 7)}{(2 \log 5 + \log 7)(\log 5 + 3 \log 7) - (\log 5 + 2 \log 7)(3 \log 5 + \log 7)}$$

$$= 5$$

7. **Ans (B)**

$$|\alpha - \beta| |\alpha + \beta| > \frac{1}{a}$$

$$\left| \frac{1}{a} \right| \left| \sqrt{\frac{1}{a^2} - 4} \right| > \frac{1}{a}$$

$$\text{As } D > 0 \Rightarrow \frac{1}{a^2} - 4 > 0$$

$$\text{If } a \in \left(-\frac{1}{2}, 0 \right) \text{ (1) is satisfied}$$

$$\text{If } a \in \left(0, \frac{1}{2} \right)$$

$$\Rightarrow \sqrt{\frac{1}{a^2} - 4} > 1 \Rightarrow \frac{-1}{\sqrt{5}} < a < \frac{1}{\sqrt{5}}$$

$$\Rightarrow \left(-\frac{1}{2}, 0 \right) \cup \left(0, \frac{1}{\sqrt{5}} \right)$$

8. **Ans (B)**

Let a_n denote the length of a side of S_n . We are given:

Length of a side of S_n = Length of a diagonal of S_{n+1}

$$\Rightarrow a_n = \sqrt{2} a_{n+1} \Rightarrow \frac{a_{n+1}}{a_n} = \frac{1}{\sqrt{2}}$$

thus, $a_1, a_2, a_3, \dots$ is a G.P. with first term 10 and common ratio $1/\sqrt{2}$. Therefore,

$$a_n = 10(1/\sqrt{2})^{n-1}$$

$$\text{Also, area}(S_n) = a_n^2 < 1$$

$$\Rightarrow [10(1/\sqrt{2})^{n-1}]^2 < 1 \Rightarrow 100 < 2^{n-1}$$

$\Rightarrow$ smallest possible value of n is 8.

9. **Ans (B)**

$$\frac{4000}{a_1 a_{4001}} = 10 \Rightarrow a_1 a_{4001} = 400$$

$$a_1 + a_{4001} = 50$$

$$(a_1 - a_{4001})^2 = (a_1 + a_{4001})^2 - 4a_1 a_{4001}$$

$$\Rightarrow |a_1 - a_{4001}| = 30$$

10. **Ans (B)**

Let $x = \tan^2 \alpha$, $y = \tan^2 \beta$

$$\therefore \frac{\sec^4 \alpha}{\tan^2 \beta} + \frac{\sec^4 \beta}{\tan^2 \alpha} = \frac{(1+x)^2}{y} + \frac{(1+y)^2}{x},$$

$x, y > 0$

$$= \left(\frac{1}{y} + 2\frac{x}{y} + \frac{x^2}{y} \right) + \left(\frac{1}{x} + \frac{2y}{x} + \frac{y^2}{x} \right)$$

$$= \left(\frac{x^2}{y} + \frac{1}{y} + \frac{y^3}{x} + \frac{1}{x} \right) + 2 \left(\frac{x}{y} + \frac{y}{x} \right) \geq 8$$

11. **Ans (B)**

$$\left(\frac{1 - 3 \tan^2 1^\circ}{3 - \tan^2 1^\circ} \right) \frac{\cot 1^\circ}{\cot 3^\circ}$$

$$= \left(\frac{1 - 3 \tan^2 1^\circ}{3 \tan 1^\circ - \tan^3 1^\circ} \right) \frac{1}{\cos 3^\circ}$$

$$= \frac{1}{\tan 3^\circ \cot 3^\circ} = 1$$

12. **Ans (B)**

$$\frac{100}{2}(x_1 + x_{100}) = -1$$

$$\frac{50}{2}(x_1 + d + x_{100}) = 1$$

13. **Ans (A)**

$$\alpha = \max \{2^{6 \sin 3x + 8 \cos 3x}\}$$

$$\text{and } \beta = \min \{2^{6 \sin 3x + 8 \cos 3x}\}$$

Now range of $6 \sin 3x + 8 \cos 3x$

$$= [-10, 10]$$

$$\alpha = 2^{10} \text{ \& } \beta = 2^{-10}$$

14. **Ans (C)**

$$x - 2y = \frac{1}{2} \left(\tan \frac{\pi}{9} + \tan \frac{7\pi}{18} \right) - \left(\tan \frac{\pi}{9} + \tan \frac{5\pi}{18} \right)$$

$$\Rightarrow |x - 2y| = \left| \frac{\cot \frac{\pi}{9} - \tan \frac{\pi}{9}}{2} - \tan \frac{5\pi}{18} \right|$$

$$= \left| \cot \frac{2\pi}{9} - \cot \frac{2\pi}{9} \right| = 0$$

15. **Ans (B)**

$$\left(\frac{1}{\sqrt{27}} \right)^{2 - (\log_5 16 / 2 \log_5 9)}$$

$$= \left(\frac{1}{\sqrt{27}} \right)^{2 - \log_2 2}$$

$$= (3)^{-3/2 (\log_3 9 / 2)}$$

$$= \left(\frac{2}{9} \right)^{3/2} = \frac{2\sqrt{2}}{27}$$

16. **Ans (C)**

$$a^2, b^2, c^2 \quad \text{AP}$$

$$\Rightarrow a^2 + 1, b^2 + 1, c^2 + 1 \quad \text{AP}$$

$$\Rightarrow a^2 + ab + bc + ca, b^2 + ab + bc + ca, c^2 + ab + bc + ca \quad \text{AP}$$

$$\Rightarrow (a+b)(a+c), (a+b)(b+c), (b+c)(c+a) \quad \text{AP}$$

$$\Rightarrow \frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b} \quad \text{AP}$$

17. **Ans (B)**

$$\text{Let } S = 1 + \frac{3}{4} + \frac{6}{4^2} + \frac{10}{4^3} + \frac{15}{4^4} + \dots$$

$$\frac{S}{4} = \frac{1}{4} + \frac{3}{4^2} + \frac{6}{4^3} + \frac{10}{4^4} + \dots$$

$$\Rightarrow \frac{3S}{4} = 1 + \frac{2}{4} + \frac{3}{4^2} + \frac{4}{4^3} + \frac{5}{4^4} + \dots$$

$$\Rightarrow \frac{3S}{16} = \frac{1}{4} + \frac{2}{4^2} + \frac{3}{4^3} + \frac{4}{4^4} + \dots$$

$$\Rightarrow \frac{9S}{16} = 1 + \frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \dots$$

$$\Rightarrow \frac{9S}{16} = \frac{1}{1 - \frac{1}{4}} \Rightarrow \frac{9S}{16} = \frac{4}{3} \Rightarrow S = \frac{64}{27}$$

18. **Ans (A)**

$$\sin \frac{\pi}{9} \left(4 + \sec \frac{\pi}{9} \right) \times \sqrt{3}$$

$$= \left(4 \sin \left(\frac{\pi}{9} \right) + \frac{\sin \pi/9}{\cos \pi/9} \right) \sqrt{3}$$

$$= \frac{\sqrt{3} \left(2 \sin \frac{2\pi}{9} + \sin \frac{\pi}{9} \right)}{\cos \frac{\pi}{9}}$$

$$= \frac{(\sin \frac{2\pi}{9} + \sin \frac{2\pi}{9} + \sin \frac{\pi}{9}) \sqrt{3}}{\cos \frac{\pi}{9}}$$

$$= \frac{\sqrt{3} \left(\cos \left(\frac{\pi}{2} - \frac{2\pi}{9} \right) \right) + \cos \frac{\pi}{18}}{\cos \frac{\pi}{9}}$$

$$= \frac{(\cos \frac{5\pi}{18} + \cos \frac{\pi}{18}) \sqrt{3}}{\cos \frac{\pi}{9}}$$

$$= \frac{\sqrt{3} \left(2 \cos \frac{\pi}{6} \cos \frac{\pi}{9} \right)}{\cos \frac{\pi}{9}}$$

$$= 2 \cdot \frac{\sqrt{3}}{2} \cdot \sqrt{3} = 3$$

19. **Ans (D)**

$$S_n = 3n^3 + 2n^2$$

$$T_6 = S_6 - S_5 = (720) - (425) = 295$$

20. **Ans (D)**

$$k - 1 \geq 0 \text{ \& } f(0) < 0$$

$$k \geq 1 \text{ \& } k^2 - 4 < 0$$

$$\Rightarrow k \in [1, 2)$$

SECTION-II

1. **Ans (2.00)**

Let α be the common root of equation

$$a\alpha^2 + b\alpha + c = 0$$

$$c\alpha^2 + b\alpha + a = 0$$

$$\therefore (a - c)\alpha^2 + (c - a) = 0$$

$$\Rightarrow \alpha = \pm 1$$

$$\therefore \alpha = -1 \text{ must be common root}$$

$$\therefore a - b + c = 0$$

2. **Ans (3.00)**

$$Q_1(x) \geq 0 \Rightarrow a > 0 \text{ and } b^2 - 4ac \leq 0$$

$$Q_2(x) \geq 0 \Rightarrow b > 0 \text{ and } c^2 - 4ab \leq 0$$

$$Q_3(x) \geq 0 \Rightarrow c > 0 \text{ and } a^2 - 4bc \leq 0$$

$$\Rightarrow \Sigma a^2 - 4(\Sigma ab) \leq 0$$

$$\Rightarrow \Sigma a^2 \leq 4(\Sigma ab) \quad \because a > 0, b > 0 \text{ and } c > 0$$

$$\therefore \Sigma ab > 0$$

$$\Rightarrow \frac{\Sigma a^2}{\Sigma ab} \leq 4 \quad \dots\dots\dots(1)$$

we known that $(a - b)^2 + (b - c)^2 + (c - a)^2 > 0$

$$\Rightarrow \Sigma a^2 - \Sigma ab > 0$$

$$\Rightarrow \frac{\Sigma a^2}{\Sigma ab} > 1 \quad \dots\dots\dots(2)$$

from (1) and (2), we can say that

$$1 < \frac{\Sigma a^2}{\Sigma ab} \leq 4$$

$\therefore$ possible of integers in the range are 2,3,4

$\therefore$ Number of integers in the range is 3.

3. **Ans (1.00)**

$$\log_{(x+1)}(2x^2 + 7x + 5) + \log_{(2x+5)}(x + 1)^2 - 4 = 0$$

$$\log_{(x+1)}(2x + 5)(x + 1) + 2\log_{(2x+5)}(x + 1) = 4$$

$$\log_{(x+1)}(2x + 5) + 1 + 2\log_{(2x+5)}(x + 1) = 4$$

$$\text{Put } \log_{(x+1)}(2x + 5) = t$$

$$t + \frac{2}{t} = 3 \Rightarrow t^2 - 3t + 2 = 0$$

$$t = 1, 2$$

$$\log_{(x+1)}(2x + 5) = 1 \text{ \& } \log_{(x+1)}(2x + 5) = 2$$

$$x + 1 = 2x + 3 \text{ \& } 2x + 5 = (x + 1)^2$$

$$x = -4 \text{ (rejected) } x^2 = 4 \Rightarrow x = 2, -2 \text{ (rejected)}$$

$$\text{so, } x = 2$$

$$\text{No. of solution} = 1$$

4. **Ans (1.00)**

$$x^2 + 5\sqrt{2}x + 10 = 0$$

$$\text{\& } p_n = \alpha^n - \beta^n \text{ (given)}$$

$$\text{Now } \frac{P_{17}P_{20} + 5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19} + 5\sqrt{2}P_{18}^2} = \frac{P_{17}(P_{20} + 5\sqrt{2}P_{19})}{P_{18}(P_{19} + 5\sqrt{2}P_{18})}$$

$$\frac{P_{17}(\alpha^{20} - \beta^{20} + 5\sqrt{2}(\alpha^{19} - \beta^{19}))}{P_{18}(\alpha^{19} - \beta^{19} + 5\sqrt{2}(\alpha^{18} - \beta^{18}))}$$

$$\frac{P_{17}(\alpha^{19}(\alpha + 5\sqrt{2}) - \beta^{19}(\beta + 5\sqrt{2}))}{P_{18}(\alpha^{18}(\alpha + 5\sqrt{2}) - \beta^{18}(\beta + 5\sqrt{2}))}$$

$$\text{Since } \alpha + 5\sqrt{2} = -10/\alpha \quad \dots\dots\dots(1)$$

$$\text{\& } \beta + 5\sqrt{2} = -10/\beta \quad \dots\dots\dots(2)$$

$$\text{Now put there values in above expression}$$

$$= -\frac{10P_{17}P_{18}}{-10P_{18}P_{17}} = 1$$

5. **Ans (3.00)**

$$\ell = \left(1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \dots \right)^{\log_{0.25} \left(\frac{1}{3} + \frac{1}{3^2} + \dots \right)}$$

$$S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \dots$$

$$\frac{S}{3} = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \dots$$

$$\frac{2S}{3} = 1 + \frac{1}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \dots$$

$$\frac{2S}{3} = \frac{4}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \dots$$

$$S = \frac{3}{2} \left(\frac{4/3}{1 - 1/3} \right) = 3$$

$$\text{Now } \ell = 3^{\log_{(1/4)} \left(\frac{1}{2} \right)} = 3^{1/2} = \sqrt{3}$$

$$\Rightarrow \ell^2 = 3$$

6. **Ans (7.00)**

We know that

$$2^{\sqrt{\log_2 3}} = 2^{(\log_2 3) \left(\frac{1}{\sqrt{\log_2 3}} \right)}$$

$$(2^{\log_2 3})^{\frac{1}{\sqrt{\log_2 3}}} = 3^{\sqrt{\log_2 3}}$$

(Using base changing formula)

$\therefore$ The given equation becomes $x^2 - 3x + 2 = 0$

$$\Rightarrow \alpha = 1, \beta = 2$$

$$\text{Hence } \alpha^2 + \alpha\beta + \beta^2 = 7$$

7. **Ans (7.00)**

$$\begin{aligned} \sum_{n=0}^{\infty} \alpha^n + \beta^n &= (1 + \alpha + \alpha^2 + \dots) + (1 + \beta + \beta^2 + \dots) \\ &= \frac{1}{1 - \alpha} + \frac{1}{1 - \beta} \\ &= \frac{2 - (\alpha + \beta)}{1 - (\alpha + \beta) + \alpha\beta} = 7 \end{aligned}$$

8. **Ans (2.00)**

$$\text{Let } P = \cos x \cos 2x \cos 3x \dots \cos 999x$$

$$\text{and } Q = \sin x \sin 2x \sin 3x \dots \sin 999x$$

$$PQ \cdot 2^{999} = \sin 2x \sin 4x \sin 6x \dots \sin 1998x$$

$$= (\sin 2x \sin 4x \dots \sin 998x) (-\sin(2\pi - 1000x))$$

$$(-\sin(2\pi - 1002x)) \dots (-\sin(2\pi - 1998x))$$

$$= \sin 2x \sin 4x \dots \sin 998x \sin 999x \sin 997x$$

$$\dots \sin x = Q$$

$$P = \frac{1}{2^{999}} (Q \neq 0)$$

9. **Ans (4.00)**

$$\frac{a+b}{2a-b} + \frac{c+b}{2c-b}$$

$$= \frac{a + \frac{2ac}{a+c}}{2c - \frac{2ac}{a+c}}$$

$$= \frac{a+3c}{2a} + \frac{3a+c}{2c}$$

$$= 1 + \frac{3}{2} \left(\frac{c}{a} + \frac{a}{c} \right) \geq 4$$

10. **Ans (4.00)**

$$x^2 - 3\sqrt{x^2} + 2 = 0$$

$$x^2 - 3|x| + 2 = 0$$

$$\text{Let } |x| = t$$

$$8t^2 - 3t + 2 = 0$$

$$t = 1, 2 \Rightarrow x = \pm 1, \pm 2$$